

How I Built a System That Catches Fraud in Under 1 Second

Every time someone pays online, there's a hidden risk. The company doesn't know right away if the card being used is real or stolen. But it has to decide — instantly — whether to accept the payment or not.

I built a system that solves exactly this problem: it checks every payment in real time and decides, in under 100 milliseconds, whether to allow it, block it, or send it to a human for a closer look.

Why this matters

When a stolen card is used, the company doesn't actually keep that money. A few weeks later, the real card owner notices the charge, tells their bank "I didn't buy this," and the bank takes the money back. By then, the company has already shipped the product — so they lose the item and the money. They may also get charged an extra fee for it. If this happens too often, the company can even lose the ability to accept card payments at all.

So catching fraud early isn't about being suspicious of customers — it's about protecting the business from a loss that's guaranteed to happen later if nothing is done.

The tricky part is speed. Checking fraud overnight, like companies used to do, is too slow — the product is already gone by the time anyone notices. The check has to happen instantly, while the customer is still on the checkout page, without making them wait or feel like they're being interrogated.

How the system works

Think of it as several quick checks happening one after another, all in the blink of an eye:

First, it looks at recent activity — has this person or card made several purchases in the last few minutes? Has this device been seen before?

Second, it checks the transaction against a list of fraud rules — like "flag anything over \$2,000 from a brand-new device." These rules can be updated instantly by the fraud team, without needing a developer to make code changes.

Third, a machine learning model looks at everything about the transaction and estimates how similar it is to past fraud cases.

All of this gets combined into one risk score. If the score is low, the payment goes through instantly. If it's high, it gets blocked. If it's somewhere in between, a real person reviews it — and whatever they decide gets remembered by the system for next time.

What makes this approach smart

No single thing proves fraud on its own. A new device isn't suspicious by itself. Multiple purchases in a few minutes isn't suspicious by itself. But when several of these things

happen together on the same transaction, that's a strong signal — and that's what the system is really looking for: patterns, not single red flags.

The system is also built to keep working even if something breaks. If the machine learning part goes down, it doesn't stop processing payments — it just relies on the simpler rules instead. A system that blocks every payment because one small part failed would be far worse than one that keeps running with slightly less intelligence.

The bigger lesson

The real challenge isn't just catching fraud — it's catching fraud without accidentally blocking real customers. Block too little, and the company loses money. Block too much, and honest customers get declined and get frustrated. The best fraud systems are the ones that stay invisible to good customers while quietly stopping the bad ones.

That balance — between protecting the business and not annoying real people — is what made this project genuinely interesting to build.

Happy to talk more about how it works if anyone's curious, or hear how others have approached similar problems.